PoseFormerV2++: Dynamic Spectral Attention and Graph-Enhanced Spatial Modeling for 3D Human Pose Estimation

Jiajie Wen[1], Mingjia Zhang[1], Le Zhang[1], Zijian Kong[1], Haifeng Li[1], and Xuejun Yue[1]*

College of Artificial Intelligence and Low-Altitude Technology, South China Agricultural University, Guangzhou 510642, China
jiajiewen@stu.scau.edu.cn, yuexuejun@scau.edu.cn

Abstract. Transformer-based methods have achieved remarkable success in sequential 2D-to-3D human pose estimation. As a pioneering transformer-based framework, PoseFormer captures spatial and temporal dependencies through cascaded transformer encoders, while PoseFormerV2 further improves efficiency and robustness through frequency-domain learning. However, PoseFormerV2 still suffers from two limitations: its frequency-domain branch relies on static spectral transformations, and its spatial modeling does not explicitly exploit skeletal topology priors. Consequently, informative motion dynamics and local structural dependencies may not be fully exploited. To address these limitations, we propose PoseFormerV2++, a transformer framework that enhances frequency-domain representation learning and topology-aware spatial modeling for 3D human pose estimation. Specifically, DSA is introduced to adaptively aggregate complementary spectral patterns from low-frequency representations, while NGRGConv explicitly incorporates skeletal topology information through topology-aware neighborhood aggregation and residual feature fusion. Extensive experiments on Human3.6M and MPI-INF-3DHP demonstrate that PoseFormerV2++ consistently outperforms PoseFormerV2 with negligible additional computational overhead, achieving a better accuracy-efficiency trade-off.

Keywords: 3D Human Pose Estimation · Transformer · Human Pose Lifting

1 Introduction

3D human pose estimation aims to recover the three-dimensional coordinates of human body joints from visual observations and serves as a fundamental task in computer vision. Accurate 3D pose estimation benefits a wide range of applications, including human-computer interaction, virtual reality, motion analysis, and intelligent surveillance. Existing approaches can be broadly categorized into

* Corresponding author.

image-based methods and lifting-based methods. Compared with image-based approaches that directly regress 3D poses from RGB images, lifting-based methods estimate 3D poses from 2D joint coordinates and have attracted increasing attention due to their lower computational cost and stronger generalization ability. [2, 13, 20, 25] With the remarkable success of transformers in sequence modeling, transformer-based lifting methods have recently achieved state-of-the-art performance by effectively capturing long-range spatial and temporal dependencies from 2D pose sequences.

Transformers [19] have achieved remarkable success in natural language processing and have subsequently become the dominant paradigm in computer vision tasks such as image classification, object detection, and video understanding. Owing to the discrete representation of human joints and the necessity of modeling long-range temporal dependencies in skeleton sequences, transformer architectures are particularly well suited for lifting-based 3D human pose estimation.

As a pioneering transformer-based framework for 3D human pose estimation, PoseFormer [30] introduces a spatial-then-temporal transformer architecture to model intra-joint and inter-frame dependencies, significantly outperforming previous convolution-based methods. Building upon PoseFormer, several studies have further improved transformer-based pose estimation. MixSTE [27] adopts alternating spatial-temporal transformer layers to better capture joint-specific motion patterns, while MHFormer [11] introduces multiple pose hypotheses to alleviate the ambiguity inherent in 2D-to-3D pose lifting. P-STMO [17] further incorporates masked joint modeling and self-supervised learning to enhance representation learning.

To further improve efficiency and robustness, PoseFormerV2 [29] improves both efficiency and robustness through frequency-domain representation learning. By transforming long skeleton sequences into compact low-frequency DCT coefficients and introducing a Time-Frequency Feature Fusion Transformer, PoseFormerV2 effectively models long-range temporal dependencies while reducing computational complexity. Figure 1 illustrates the overall architecture of PoseFormerV2.

Despite its strong performance, PoseFormerV2 still exhibits two limitations. First, in the frequency-domain branch, PoseFormerV2 employs a standard MLP to process low-frequency DCT representations. Although this design is simple and effective, it treats different spectral patterns with a uniform feed-forward transformation and lacks an adaptive mechanism to emphasize informative spectral components according to input motion dynamics. Consequently, discriminative motion cues contained in low-frequency representations may not be fully exploited. Second, the spatial branch primarily relies on self-attention to model inter-joint relationships without explicitly incorporating skeletal topology priors. Since physically connected joints exhibit strong kinematic dependencies, the absence of topology-aware modeling may limit local structural reasoning, particularly under noisy observations and complex motions.

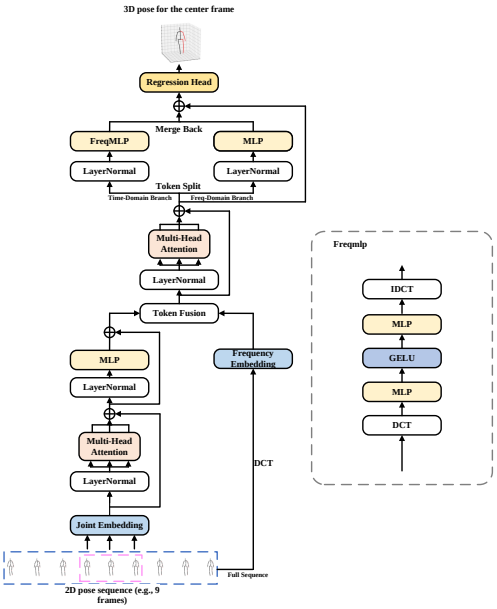


Fig. 1: Overview of PoseFormerV2. The framework consists of a Spatial Transformer and a Time-Frequency Feature Fusion Transformer. The time-domain tokens are processed by FreqMLP, whereas the frequency-domain tokens are processed by a vanilla MLP. The two branches are subsequently fused for efficient temporal modeling.

Motivated by the above observations, we argue that further improvements of PoseFormerV2 require enhancements in both temporal spectral representation learning and spatial structural modeling. Specifically, the frequency-domain branch should be able to adaptively emphasize informative motion patterns rather than relying on uniform feature transformations [7], while the spatial modeling process should explicitly exploit skeletal topology priors beyond self-attention alone [21].

In this paper, we present PoseFormerV2++, a simple yet effective extension of PoseFormerV2. For temporal modeling, we introduce Dynamic Spectral Attention (DSA) to adaptively aggregate complementary spectral patterns from low-frequency representations, enabling the model to selectively focus on informative motion dynamics. For spatial modeling, we propose Node-Gated Residual Graph Convolution (NGRGConv) to explicitly incorporate skeletal topology information through topology-aware neighborhood aggregation and adaptive feature fusion. By jointly enhancing frequency-domain representation learning and spatial structural reasoning, the proposed framework achieves consistent performance gains while introducing almost no additional computational cost.

– We propose DSA to enhance frequency-domain representation learning by adaptively aggregating spectral patterns.

- We introduce NGRGConv to explicitly inject skeletal topology priors into spatial modeling.
- Extensive experiments on Human3.6M [6] and MPI-INF-3DHP [14] demonstrate that PoseFormerV2++ consistently improves PoseFormerV2 with negligible computational overhead.

2 Related Work

Our method is built upon PoseFormerV2 [29], a transformer-based framework that achieves efficient long-range temporal modeling through frequency-domain representations and time-frequency feature fusion. Therefore, this section reviews related works on transformer-based 3D human pose estimation, spectral representation learning, and graph-based skeleton modeling, which are most relevant to our proposed approach.

2.1 Transformer-based 3D Human Pose Estimation

Transformer-based methods have become the dominant paradigm for lifting-based 3D human pose estimation due to their strong capability in modeling long-range spatial and temporal dependencies. PoseFormer [30] is the first work to introduce transformer architectures into 2D-to-3D pose lifting and significantly outperforms previous CNN-based approaches. Building upon PoseFormer, several studies have explored more effective spatio-temporal modeling strategies. MixSTE [27] argues that the original spatial-then-temporal paradigm may overlook joint-specific temporal dynamics and proposes alternating spatial-temporal transformer layers for fine-grained motion modeling. MHFormer [11] further introduces multiple pose hypotheses to address the inherent ambiguity of 2D-to-3D lifting and achieves improved estimation accuracy. Subsequent works such as P-STMO [17] incorporate masked joint modeling and self-supervised learning to enhance representation learning.

PoseFormerV2 [29] further improves transformer-based 3D human pose estimation from a frequency-domain perspective. By representing long sequences with compact low-frequency DCT coefficients and introducing a Time-Frequency Feature Fusion Transformer, PoseFormerV2 achieves efficient long-range temporal modeling while maintaining strong robustness to noisy 2D joint detections. However, its temporal branch still relies on static feed-forward transformations for spectral representation learning, and its spatial modeling does not explicitly exploit skeletal topology priors. Our work further improves PoseFormerV2 through adaptive spectral representation learning and topology-aware spatial modeling.

2.2 Spectral Representation Learning

Frequency-domain representations have been widely used for modeling long-range temporal dynamics in skeleton sequences [23, 31, 32]. PoseFormerV2 [29]

introduces low-frequency DCT representations and time-frequency feature fusion into lifting-based 3D human pose estimation, demonstrating the effectiveness of spectral modeling for efficient long-sequence learning. More recently, spectral attention mechanisms [7] have shown superior capability in adaptively aggregating complementary spectral patterns. Our DSA further extends this idea for frequency-domain representation learning in 3D human pose estimation.

2.3 Graph-based Skeleton Modeling

Graph neural networks [1, 4, 22, 26, 28, 33] have been widely adopted for skeleton-based human motion analysis due to their ability to explicitly model the kinematic relationships among body joints. Early works such as ST-GCN [22] construct human skeletons as graphs and perform message passing according to predefined physical connections. Subsequent methods, including 2s-AGCN [18] and CTR-GCN [4], further improve graph modeling by introducing adaptive graph structures and channel-wise topology refinement, enabling more flexible joint dependency learning.

Furthermore, graph-based modules have been integrated with transformer architectures to combine local structural reasoning and global dependency modeling. HandGCNFormer [21] introduces node-offset graph convolution to strengthen topology-aware feature extraction before self-attention. However, most transformer-based 3D human pose estimation methods, including PoseFormerV2 [29], primarily rely on self-attention for spatial modeling and do not explicitly exploit skeletal topology priors. Motivated by these observations, we introduce NGRGConv to incorporate topology-aware structural information into PoseFormerV2.

3 Method

PoseFormerV2 achieves efficient long-range temporal modeling through frequency-domain representations and time-frequency feature fusion. However, it does not explicitly exploit skeletal topology priors, and its frequency-domain branch relies on a standard feed-forward network to model low-frequency spectral representations. Such a design lacks an adaptive mechanism for exploiting complementary motion patterns encoded at different temporal scales. To address these limitations, we propose PoseFormerV2++, which introduces graph-enhanced spatial modeling and Dynamic Spectral Attention to jointly improve topology-aware structural reasoning and adaptive spectral representation learning.

3.1 Preliminary of PoseFormerV2

Given an input 2D skeleton sequence

$$\mathbf{x} \in \mathbb{R}^{F \times J \times 2}, \tag{1}$$

where F and J denote the sequence length and the number of body joints, respectively. PoseFormerV2 first samples F' frames around the sequence center,

denoted by $\mathbf{x}' \in \mathbb{R}^{F' \times J \times 2}$, and feeds them into a spatial transformer encoder to model intra-frame joint dependencies. The output of the spatial encoder is represented as

$$\mathbf{z}^{Time} \in \mathbb{R}^{F' \times (J \cdot c)}. \tag{2}$$

To capture long-range temporal dynamics, the full sequence $\mathbf{x}$ is transformed into the frequency domain using Discrete Cosine Transform (DCT). The first N low-frequency coefficients are retained and linearly projected into frequency embeddings:

$$\mathbf{z}^{Freq} \in \mathbb{R}^{N \times (J \cdot c)}. \tag{3}$$

The time-domain and frequency-domain features are concatenated as

$$\mathbf{z} = [\mathbf{z}^{Time}; \mathbf{z}^{Freq}], \tag{4}$$

and fed into a Time-Frequency Feature Fusion Transformer for temporal dependency modeling.

Specifically, the time-domain and frequency-domain tokens share the same multi-head self-attention (MSA) layer while employing separate feed-forward networks. For the time-domain branch, PoseFormerV2 introduces a Frequency Multi-Layer Perceptron (FreqMLP), which applies DCT and inverse DCT (IDCT) before and after a vanilla MLP, respectively, to perform learnable frequency-domain filtering:

$$\mathbf{z}'_k = MSA(\mathbf{z}_k), \tag{5}$$

$$\mathbf{z}_k^{Time}, \mathbf{z}_k^{Freq} = \mathbf{z}'_k[:F'], \mathbf{z}'_k[F':], \tag{6}$$

$$\mathbf{z}_{k+1} = Concat\left(FreqMLP(\mathbf{z}_k^{Time}), MLP(\mathbf{z}_k^{Freq})\right). \tag{7}$$

Finally, a regression head consisting of a 1D convolution layer and a linear projection is adopted to predict the 3D pose of the target frame:

$$\mathbf{y} \in \mathbb{R}^{1 \times (J \cdot 3)}. \tag{8}$$

Limitation of PoseFormerV2. PoseFormerV2 improves the efficiency and robustness of transformer-based 3D human pose estimation by introducing a compact frequency-domain representation and a time-frequency fusion strategy. Nevertheless, two limitations remain. First, in the frequency-domain branch, PoseFormerV2 employs a standard feed-forward network to process low-frequency spectral representations. Although effective, such a design treats all spectral patterns uniformly and lacks an adaptive mechanism for emphasizing informative motion dynamics. Consequently, complementary spectral cues encoded at different temporal scales may not be fully exploited. Second, the spatial transformer models inter-joint relationships solely through self-attention while overlooking

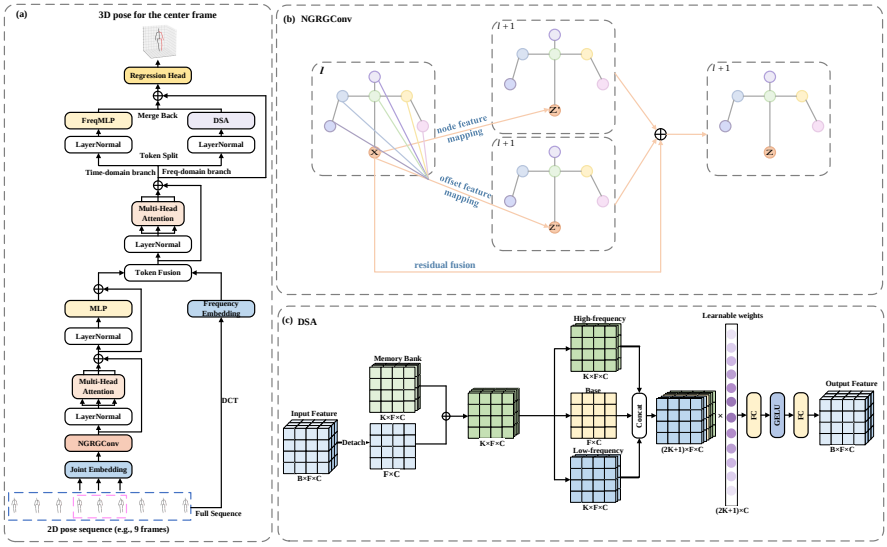


Fig. 2: (a) Overview of PoseFormerV2++. (b) Node-Gated Residual Graph Convolution (NGRGConv). (c) Dynamic Spectral Attention (DSA). PoseFormerV2++ enhances the original PoseFormerV2 from both spatial and temporal perspectives. Specifically, NGRGConv explicitly incorporates skeletal topology priors through topology-aware neighborhood aggregation and residual feature fusion before the spatial transformer, while DSA adaptively aggregates complementary spectral patterns from low-frequency DCT representations in the frequency-domain branch. The enhanced spatial and frequency-domain representations are subsequently fused for efficient long-range temporal modeling and accurate 3D human pose estimation.

the inherent skeletal topology prior. Since physically connected joints exhibit strong kinematic dependencies, the absence of explicit topology-aware modeling may limit the preservation of local structural information, especially under noisy observations and complex human motions.

3.2 PoseFormerV2++

Motivation To address the above limitations, we propose PoseFormerV2++, which enhances both temporal representation learning and spatial structural reasoning. The overall architecture of PoseFormerV2++ is illustrated in Fig. 2. For temporal modeling, we introduce a Dynamic Spectral Attention (DSA) module in the frequency-domain branch to adaptively aggregate complementary spectral patterns extracted from low-frequency DCT representations. Instead of relying on a static feed-forward transformation, DSA constructs and aggregates input-dependent spectral components from the current frequency-domain features and the running spectral memory, enabling more discriminative frequency-domain representation learning. For spatial modeling, we design a Node-Gated Residual

Graph Convolution (NGRGConv) module to explicitly inject skeletal topology priors before spatial self-attention. By adaptively fusing node-wise representations and topology-aware neighborhood features, NGRGConv strengthens local kinematic reasoning while preserving the global modeling capability of transformers. Through the joint enhancement of adaptive spectral representation learning and topology-aware spatial modeling, PoseFormerV2++ produces more discriminative spatio-temporal features for 3D human pose estimation.

Dynamic Spectral Attention (DSA) In PoseFormerV2, the frequency-domain branch models low-frequency DCT representations using a standard feed-forward network. Although effective, such a design treats all spectral patterns uniformly and lacks an explicit mechanism for modeling complementary spectral components. To address this limitation, we propose DSA, which constructs and aggregates complementary spectral components over frequency-domain representations. Our design is related to recent spectral attention mechanisms for adaptive frequency representation learning [7].

Given the frequency-domain feature sequence

$$\mathbf{Z}^{Freq} \in \mathbb{R}^{N \times C}, \tag{9}$$

DSA first constructs a set of complementary spectral components through multiple exponential moving average (EMA) filters with different smoothing factors. Specifically, the low-frequency components are computed as

$$\mathbf{L}_i = \alpha_i \mathbf{M}_i + (1 - \alpha_i)\mathbf{Z}^{Freq}, \tag{10}$$

where $\mathbf{M}_i$ denotes the running spectral memory. After each update, the resulting spectral memory is detached from the computation graph and retained for subsequent feature construction. In all experiments, we set $K = 4$ and use four independently learnable smoothing coefficients. Each coefficient is parameterized as $\alpha_i = \sigma(\theta_i)$ to constrain it to $(0, 1)$ and is initialized as $\{\alpha_i\}_{i=1}^{4} = \{0.5, 0.7, 0.9, 0.99\}$. Each DSA block maintains its own set of smoothing coefficients. Correspondingly, the high-frequency residual components are obtained by

$$\mathbf{H}_i = \mathbf{Z}^{Freq} - \mathbf{L}_i. \tag{11}$$

Together with the original spectral representation, these components form a spectral component bank:

$$\mathcal{B} = \left\{\mathbf{H}_1, \ldots, \mathbf{H}_K, \mathbf{Z}^{Freq}, \mathbf{L}_1, \ldots, \mathbf{L}_K\right\}. \tag{12}$$

Here, "dynamic" refers to the input-dependent spectral components constructed from the current frequency-domain features and the running spectral memory.

To exploit motion patterns encoded at different spectral scales, DSA learns channel-wise attention weights over the component bank:

$$\mathbf{W} = \mathrm{Softmax}(\mathbf{S}), \tag{13}$$

where $\mathbf{S}$ is a learnable spectral attention parameter. The final aggregated representation is computed as

$$\widetilde{\mathbf{Z}}^{Freq} = \sum_{m=1}^{2K+1} \mathbf{W}_m \odot \mathcal{B}_m. \tag{14}$$

Finally, a lightweight feed-forward network is applied to further enhance the aggregated spectral representation. By aggregating complementary input-dependent spectral components rather than relying on static feature transformations, DSA provides a more discriminative frequency-domain representation for subsequent time-frequency fusion.

Node-Gated Residual Graph Convolution (NGRGConv) PoseFormerV2 relies entirely on self-attention to model spatial dependencies among joints. Although self-attention is effective for capturing global correlations, it does not explicitly exploit the inherent skeletal topology prior. Physically connected joints exhibit strong kinematic dependencies, which are particularly important under noisy observations and complex motions. To explicitly incorporate skeletal topology information before spatial self-attention, we introduce NGRGConv. The proposed design is closely related to topology-aware graph convolution methods such as HandGCNFormer [21].

Given the spatial token representation

$$\mathbf{X} \in \mathbb{R}^{J \times C}, \tag{15}$$

where J denotes the number of joints and C denotes the feature dimension. We construct a fixed undirected skeleton graph $G = (V, E)$ according to the physical connections of the 17-joint Human3.6M skeleton. Self-loops are added to the adjacency matrix before symmetric normalization. The normalized adjacency matrix is computed as

$$\hat{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-\frac{1}{2}}, \tag{16}$$

where $\mathbf{A}$ is the adjacency matrix, $\mathbf{I}$ denotes the identity matrix, and $\mathbf{D}$ is the degree matrix of $\mathbf{A} + \mathbf{I}$.

Following the intuition of node-offset graph convolution, we decouple node-wise feature transformation and topology-aware neighborhood aggregation into two parallel branches. The node feature branch preserves the representation of each joint individually:

$$\mathbf{F}_{node} = \mathbf{X}\mathbf{W}_1, \tag{17}$$

while the neighborhood branch aggregates information from connected joints through graph propagation:

$$\mathbf{F}_{offset} = \left(\hat{\mathbf{A}}\mathbf{X}\right)\mathbf{W}_2, \tag{18}$$

To adaptively balance the contributions of local node information and neighborhood structural cues, two learnable gating coefficients are introduced:

$$\mathbf{F} = \gamma_1 \mathbf{F}_{node} + \gamma_2 \mathbf{F}_{offset}, \tag{19}$$

where γ_1 and γ_2 are trainable scalar parameters, both initialized to zero.

Finally, a residual connection is employed to preserve the original joint representation and facilitate optimization:

$$\mathbf{Y} = \mathbf{X} + \sigma(\mathbf{F}), \tag{20}$$

where $\sigma(\cdot)$ denotes the GELU activation function.

By explicitly injecting skeletal topology priors before the spatial transformer encoder, NGRGConv strengthens local kinematic reasoning while maintaining compatibility with the subsequent self-attention mechanism.

4 Experiments

4.1 Datasets and Evaluation Metrics

We evaluate the proposed method on two widely used benchmarks for 3D human pose estimation, including Human3.6M [6] and MPI-INF-3DHP [14].

Human3.6M is the most commonly used indoor benchmark for 3D human pose estimation, containing videos captured from four synchronized cameras under multiple action categories. Following previous works, subjects S1, S5, S6, S7, and S8 are used for training, while S9 and S11 are adopted for evaluation.

MPI-INF-3DHP contains both indoor and outdoor scenes with more challenging motion variations and camera viewpoints. Similar to Human3.6M, it also provides multi-view recordings under different subjects and action categories.

Evaluation Metrics. For Human3.6M, we report MPJPE (Mean Per Joint Position Error) and P-MPJPE following standard evaluation protocols. MPJPE computes the average Euclidean distance between predicted joints and ground-truth joints, while P-MPJPE further aligns the predicted pose with the ground truth via rigid transformation. For MPI-INF-3DHP, we additionally report PCK (Percentage of Correct Keypoint) and AUC (Area Under Curve).

4.2 Implementation Details

Following PoseFormerV2, we adopt three hyper-parameters in our experiments, including the number of frame-kept tokens f, the input sequence length F, and the number of retained DCT coefficients n. Unless otherwise specified, we set $n = f$. Different settings of f, F, and n provide flexible trade-offs between computational efficiency and pose estimation accuracy.

Our implementation is built upon PoseFormerV2 and follows the original training settings unless otherwise specified. On Human3.6M, we use CPN-detected 2D poses as inputs. For MPI-INF-3DHP, we use ground-truth 2D poses following

Table 1: Comparison with existing methods on Human3.6M. All results are obtained using CPN-detected 2D poses as inputs. The best results are highlighted in bold.

Method	Frame Sequence Length	MFLOPs	MPJPE ↓	P-MPJPE ↓
StridedTrans [10]TMM'22	81	81	342.5	47.5	–
MixSTE [27]CVPR'22	3	3	3420	49.6	38.9
MHFormer [11]CVPR'22	9	9	342.9	47.8	–
MHFormer [11]CVPR'22	27	27	1031.8	45.9	–
P-STMO [17] ECCV'22	27	81	163.0	46.8	–
P-STMO [17] ECCV'22	81	81	493.0	45.6	–
Einfalt *et al.* [5]WACV'23	9	81	543.0	47.9	–
DC-GCT [8]TMM'25	9	9	77.0	46.5	–
PoseSSM [24]AAAI'25	1	1	35.6	48.1	37.9
SCJD [3]ICME'25	9	81	81.3	46.3	–
PoseFormerV1 [30]ICCV'21	27	27	542.1	47.0	–
PoseFormerV2 [29]CVPR'23	3	27	117.3	47.9	37.4
PoseFormerV2 [29]CVPR'23	9	81	351.7	46.0	36.1
PoseFormerV2 [29]CVPR'23	27	243	1054.8	**45.2**	**35.6**
PoseFormerV2++(Ours)	3	27	117.8	46.0	36.6
PoseFormerV2++(Ours)	9	81	353.1	45.4	35.9

PoseFormerV2. Horizontal flipping is adopted during both training and testing. No additional pre-training or refinement module is used.

Experiments are conducted using PyTorch on a single NVIDIA RTX 3090 GPU with an Intel i9-10900K CPU and 64GB memory. We train the model using AdamW for 80 epochs with an initial learning rate of 8×10^{-4} and a weight decay of 0.1. The learning rate is decayed exponentially with a decay factor of 0.99. All other training and architectural hyper-parameters are kept the same as PoseFormerV2 for fair comparison.

4.3 Comparison with Existing Methods

Table 1, presents the quantitative comparison on the Human3.6M dataset. Compared with PoseFormerV2, the proposed PoseFormerV2++ consistently achieves lower MPJPE across different temporal settings while introducing only negligible computational overhead. Under the 27-frame setting, our method improves MPJPE from 47.9 mm to 46.0 mm with merely 0.5 additional MFLOPs. Under the 81-frame setting, PoseFormerV2++ further reduces MPJPE from 46.0 mm to 45.4 mm while increasing the computational cost by only 1.4 MFLOPs. Additionally, three independent runs under 81/9/9 yield 45.4, 45.4, and 45.5 mm MPJPE (mean 45.43 ± 0.05 mm). These results demonstrate that the proposed DSA and NGRGConv modules effectively enhance temporal and spatial representation learning without sacrificing the computational efficiency inherited from PoseFormerV2.

Table 2: Comparison with existing methods on MPI-INF-3DHP. The best results are highlighted in bold.

Method	Publication	PCK ↑	AUC ↑	MPJPE ↓
Mehta *et al.* [14]	3DV'2017	75.7	39.3	117.6
Mehta *et al.* [15]	ACM ToG'2017	76.6	40.4	124.7
Pavllo *et al.* [16] (T=81)	CVPR'2019	86.0	51.9	84.0
Pavllo *et al.* [16] (T=243)	CVPR'2019	85.5	51.5	84.8
Lin *et al.* [12] (T=25)	BMVC'2019	83.6	51.4	79.8
Li *et al.* [9]	CVPR'2020	81.2	46.1	99.7
Chen *et al.* [2] (T=81)	TCSVT'2021	87.9	54.0	78.8
PoseFormerV1 [30] (T=9)	ICCV'2021	95.4	63.2	57.7
MHFormer [11] (T=9)	CVPR'2022	93.8	63.3	58.0
MixSTE [27] (T=27)	CVPR'2022	94.4	66.5	54.9
P-STMO [17] (T=81)	ECCV'2022	97.9	75.8	32.2
DC-GCT [8]	TMM'2025	87.5	55.9	–
PoseSSM [24]	AAAI'2025	86.7	54.5	–
SCJD [3]	ICME'2025	97.5	76.5	30.7
PoseFormerV2 [29] (T=81)	CVPR'2023	97.9	78.8	27.8
PoseFormerV2++(Ours) (T=81)		**98.3**	**82.6**	**27.2**

The results on MPI-INF-3DHP are reported in Table 2. Compared with PoseFormerV2, PoseFormerV2++ improves all three evaluation metrics. In particular, our method improves the AUC score from 78.8 to 82.6, indicating a more consistent reduction of pose estimation errors over different distance thresholds. We attribute this improvement to the enhanced robustness of the proposed spectral modeling and topology-aware spatial reasoning. The improvements in PCK, AUC, and MPJPE further demonstrate the effectiveness of the proposed framework on MPI-INF-3DHP.

Qualitative comparisons on Human3.6M are shown in Fig. 3. Compared with PoseFormerV2 and PoseFormerV1, the proposed method produces predictions that are visually closer to the ground truth, particularly for challenging limb articulations. More accurate localization can be observed around hand joints, wrists, knees, and ankles, where subtle pose variations are difficult to estimate. Moreover, the predicted poses exhibit better structural consistency and preserve fine-grained body configurations more faithfully. These observations further validate the effectiveness of the proposed DSA and NGRGConv modules in improving both temporal motion modeling and spatial structural representation.

4.4 Ablation Study

Table 3 presents the ablation study of different DSA insertion strategies and the proposed NGRGConv module. Replacing the standard MLP in the frequency-domain branch with DSA yields the largest individual improvement, reducing MPJPE from 47.9 mm to 46.2 mm. This result indicates that adaptive spectral

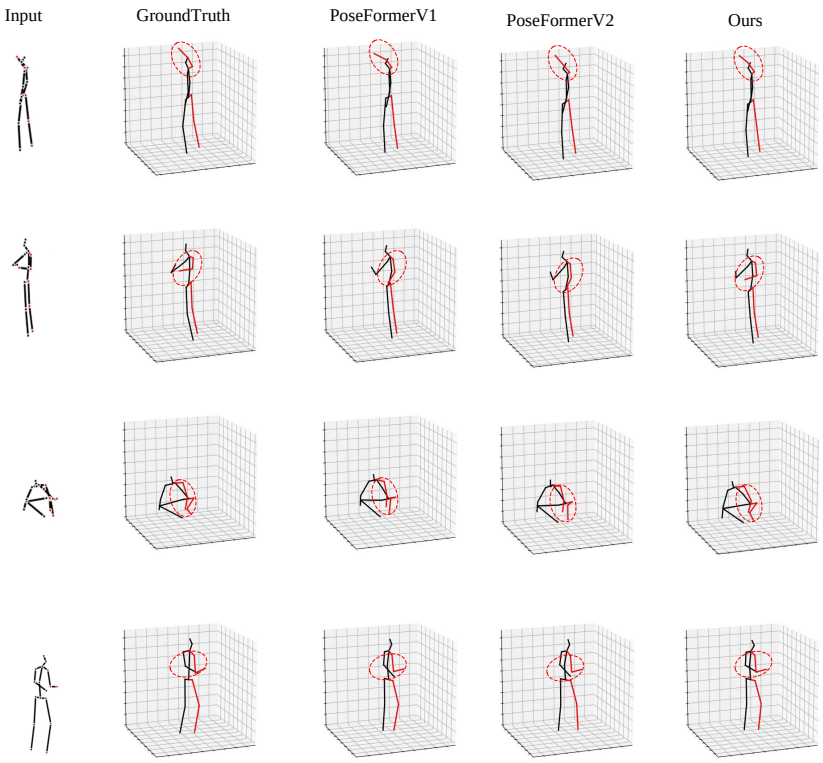


Fig. 3: Qualitative comparison on Human3.6M. Compared with PoseFormerV1 and PoseFormerV2, PoseFormerV2++ produces predictions that are visually closer to the ground truth, particularly for subtle limb articulations. The red circles highlight representative regions where our method achieves more accurate joint reconstruction.

aggregation is particularly effective for modeling low-frequency DCT representations. In contrast, replacing the internal MLP within FreqMLP achieves a smaller gain, suggesting that the original FreqMLP already provides effective frequency-domain filtering through the DCT-IDCT transformation. Applying DSA to both branches further improves the baseline but does not outperform the frequency-branch-only variant, implying that excessive spectral modulation may introduce redundant feature transformations. Meanwhile, NGRGConv consistently improves performance by explicitly incorporating skeletal topology priors into spatial feature extraction. Finally, combining frequency-branch DSA and NGRGConv achieves the best result of 46.0 mm with only a marginal increase in computational cost, demonstrating the complementary nature of adaptive spectral modeling and topology-aware spatial reasoning.

Table 3: Ablation study of different components under the 27-frame setting with 3 frame-kept and 3 coefficient-kept tokens on Human3.6M.

Step	Description	MFLOPs	MPJPE↓
(0)	Original PoseFormerV2	117.3	47.9
(1)	Replace frequency-branch MLP with DSA	117.3	46.2
(2)	Replace internal MLP in FreqMLP with DSA	117.3	46.9
(3)	Apply DSA to both branches	117.3	46.8
(4)	Add NGRGConv	117.8	46.7
(5)	Frequency-Branch DSA & NGRGConv (Full Model)	117.8	**46.0**

Table 4: Learned parameters of the best 81/9/9 checkpoint.

Block	Learned $\{\alpha_i\}_{i=1}^{4}$	High/Base/Low (%)
1	$\{0.602, 0.697, 0.768, 0.844\}$	43.6/15.4/41.0
2	$\{0.692, 0.764, 0.829, 0.887\}$	50.1/13.9/36.1
3	$\{0.714, 0.800, 0.858, 0.905\}$	51.1/13.3/35.6
4	$\{0.739, 0.826, 0.855, 0.899\}$	49.5/13.8/36.8
NGRGConv gates: $\gamma_1 = 0.040$, $\gamma_2 = 0.195$		

As shown in Table 4, all spectral groups remain active, while deeper DSA blocks assign relatively larger attention mass to high-frequency residual components. The larger learned value of γ_2 indicates a stronger gate on topology-aware neighborhood aggregation.

5 Conclusion

In this paper, we presented PoseFormerV2++, an enhanced transformer framework for 3D human pose estimation. Specifically, we introduced DSA to improve adaptive spectral representation learning in the frequency domain and NGRGConv to explicitly incorporate skeletal topology priors into spatial modeling. By jointly leveraging DSA and NGRGConv, the proposed framework produces more discriminative spatio-temporal representations while introducing negligible computational overhead. Extensive experiments on Human3.6M and MPI-INF-3DHP demonstrate that PoseFormerV2++ consistently outperforms PoseFormerV2 and achieves competitive performance against existing methods. Future work will investigate more adaptive frequency modeling strategies and topology-aware architectures for robust 3D human pose estimation.

References

1. Cai, Y., Ge, L., Liu, J., Cai, J., Cham, T.J., Yuan, J., Thalmann, N.M.: Exploiting spatial-temporal relationships for 3d pose estimation via graph convolutional

networks. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 2272–2281 (2019)
2. Chen, T., Fang, C., Shen, X., Zhu, Y., Chen, Z., Luo, J.: Anatomy-aware 3d human pose estimation with bone-based pose decomposition. IEEE Transactions on Circuits and Systems for Video Technology **32**(1), 198–209 (2021)
3. Chen, W., Xu, X., Yang, H., Xie, Y., Xiao, P., Xu, C., Zhang, H., Heng, P.A.: Scjd: Sparse correlation and joint distillation for efficient 3d human pose estimation. In: 2025 IEEE International Conference on Multimedia and Expo (ICME). pp. 1–6. IEEE (2025)
4. Chen, Y., Zhang, Z., Yuan, C., Li, B., Deng, Y., Hu, W.: Channel-wise topology refinement graph convolution for skeleton-based action recognition. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 13359–13368 (2021)
5. Einfalt, M., Ludwig, K., Lienhart, R.: Uplift and upsample: Efficient 3d human pose estimation with uplifting transformers. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision. pp. 2903–2913 (2023)
6. Ionescu, C., Papava, D., Olaru, V., Sminchisescu, C.: Human3. 6m: Large scale datasets and predictive methods for 3d human sensing in natural environments. IEEE transactions on pattern analysis and machine intelligence **36**(7), 1325–1339 (2013)
7. Kang, B.G., Lee, D., Kim, H., Chung, D., Yoon, S.: Introducing spectral attention for long-range dependency in time series forecasting. Advances in Neural Information Processing Systems **37**, 136509–136544 (2024)
8. Kang, H., Wang, Y., Liu, M., Wu, D., Liu, P., Yang, W.: Double-chain graph convolution transformer for 3d human pose estimation. IEEE Transactions on Multimedia (2025)
9. Li, S., Ke, L., Pratama, K., Tai, Y.W., Tang, C.K., Cheng, K.T.: Cascaded deep monocular 3d human pose estimation with evolutionary training data. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 6173–6183 (2020)
10. Li, W., Liu, H., Ding, R., Liu, M., Wang, P., Yang, W.: Exploiting temporal contexts with strided transformer for 3d human pose estimation. IEEE Transactions on Multimedia **25**, 1282–1293 (2022)
11. Li, W., Liu, H., Tang, H., Wang, P., Van Gool, L.: Mhformer: Multi-hypothesis transformer for 3d human pose estimation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 13147–13156 (2022)
12. Lin, J., Lee, G.H.: Trajectory space factorization for deep video-based 3d human pose estimation. arXiv preprint arXiv:1908.08289 (2019)
13. Liu, R., Shen, J., Wang, H., Chen, C., Cheung, S.c., Asari, V.: Attention mechanism exploits temporal contexts: Real-time 3d human pose reconstruction. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 5064–5073 (2020)
14. Mehta, D., Rhodin, H., Casas, D., Fua, P., Sotnychenko, O., Xu, W., Theobalt, C.: Monocular 3d human pose estimation in the wild using improved cnn supervision. In: 2017 international conference on 3D vision (3DV). pp. 506–516. IEEE (2017)
15. Mehta, D., Sridhar, S., Sotnychenko, O., Rhodin, H., Shafiei, M., Seidel, H.P., Xu, W., Casas, D., Theobalt, C.: Vnect: Real-time 3d human pose estimation with a single rgb camera. Acm transactions on graphics (tog) **36**(4), 1–14 (2017)
16. Pavllo, D., Feichtenhofer, C., Grangier, D., Auli, M.: 3d human pose estimation in video with temporal convolutions and semi-supervised training. In: Proceedings of

the IEEE/CVF conference on computer vision and pattern recognition. pp. 7753–7762 (2019)
17. Shan, W., Liu, Z., Zhang, X., Wang, S., Ma, S., Gao, W.: P-stmo: Pre-trained spatial temporal many-to-one model for 3d human pose estimation. In: European Conference on Computer Vision. pp. 461–478. Springer (2022)
18. Shi, L., Zhang, Y., Cheng, J., Lu, H.: Two-stream adaptive graph convolutional networks for skeleton-based action recognition. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 12026–12035 (2019)
19. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. Advances in neural information processing systems **30** (2017)
20. Wang, J., Yan, S., Xiong, Y., Lin, D.: Motion guided 3d pose estimation from videos. In: European conference on computer vision. pp. 764–780. Springer (2020)
21. Wang, Y., Chen, L., Li, J., Zhang, X.: Handgcnformer: A novel topology-aware transformer network for 3d hand pose estimation. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. pp. 5675–5684 (2023)
22. Yan, S., Xiong, Y., Lin, D.: Spatial temporal graph convolutional networks for skeleton-based action recognition. In: Proceedings of the AAAI conference on artificial intelligence. vol. 32 (2018)
23. Yi, K., Zhang, Q., Fan, W., Wang, S., Wang, P., He, H., An, N., Lian, D., Cao, L., Niu, Z.: Frequency-domain mlps are more effective learners in time series forecasting. Advances in Neural Information Processing Systems **36**, 76656–76679 (2023)
24. Yin, H., You, S., Han, J., Chen, Z.: Sequential joint dependency aware human pose estimation with state space model. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 39, pp. 9499–9507 (2025)
25. Zeng, A., Sun, X., Huang, F., Liu, M., Xu, Q., Lin, S.: Srnet: Improving generalization in 3d human pose estimation with a split-and-recombine approach. In: European Conference on Computer Vision. pp. 507–523. Springer (2020)
26. Zeng, A., Sun, X., Yang, L., Zhao, N., Liu, M., Xu, Q.: Learning skeletal graph neural networks for hard 3d pose estimation. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 11436–11445 (2021)
27. Zhang, J., Tu, Z., Yang, J., Chen, Y., Yuan, J.: Mixste: Seq2seq mixed spatio-temporal encoder for 3d human pose estimation in video. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 13232–13242 (2022)
28. Zhao, L., Peng, X., Tian, Y., Kapadia, M., Metaxas, D.N.: Semantic graph convolutional networks for 3d human pose regression. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 3425–3435 (2019)
29. Zhao, Q., Zheng, C., Liu, M., Wang, P., Chen, C.: Poseformerv2: Exploring frequency domain for efficient and robust 3d human pose estimation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 8877–8886 (2023)
30. Zheng, C., Zhu, S., Mendieta, M., Yang, T., Chen, C., Ding, Z.: 3d human pose estimation with spatial and temporal transformers. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 11656–11665 (2021)
31. Zhou, T., Ma, Z., Wen, Q., Wang, X., Sun, L., Jin, R.: Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In: International conference on machine learning. pp. 27268–27286. PMLR (2022)

32. Zhu, W., Ma, X., Liu, Z., Liu, L., Wu, W., Wang, Y.: Motionbert: A unified perspective on learning human motion representations. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 15085–15099 (2023)
33. Zou, Z., Tang, W.: Modulated graph convolutional network for 3d human pose estimation. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 11477–11487 (2021)